

Writing Logs in Terms of Others

Date_____ Period____

Use the properties of logarithms and the values below to find the logarithm indicated. Do not use a calculator to evaluate the logs.

1) $\log 12 \approx 1.1$
 $\log 7 \approx 0.8$
 $\log 8 \approx 0.9$
Find $\log \frac{1}{49}$

2) $\log 12 \approx 1.1$
 $\log 8 \approx 0.9$
 $\log 7 \approx 0.8$
Find $\log \frac{1}{7}$

3) $\log 12 \approx 1.1$
 $\log 7 \approx 0.8$
 $\log 8 \approx 0.9$
Find $\log \frac{1}{12}$

4) $\log 8 \approx 0.9$
 $\log 7 \approx 0.8$
 $\log 12 \approx 1.1$
Find $\log \frac{3}{2}$

5) $\log 7 \approx 0.8$
 $\log 12 \approx 1.1$
 $\log 8 \approx 0.9$
Find $\log \frac{2}{3}$

6) $\log 7 \approx 0.8$
 $\log 12 \approx 1.1$
 $\log 8 \approx 0.9$
Find $\log 64$

7) $\log 7 \approx 0.8$
 $\log 12 \approx 1.1$
 $\log 8 \approx 0.9$
Find $\log \frac{7}{12}$

8) $\log 7 \approx 0.8$
 $\log 12 \approx 1.1$
 $\log 8 \approx 0.9$
Find $\log 144$

9) $\log_7 11 \approx 1.2$
 $\log_7 6 \approx 0.9$
 $\log_7 4 \approx 0.7$
Find $\log_7 \frac{24}{11}$

10) $\log_7 6 \approx 0.9$
 $\log_7 9 \approx 1.1$
 $\log_7 11 \approx 1.2$
Find $\log_7 \frac{14}{3}$

11) $\log_3 10 \approx 2.1$
 $\log_3 8 \approx 1.9$
 $\log_3 7 \approx 1.8$
Find $\log_3 700$

12) $\log_9 6 \approx 0.8$
 $\log_9 8 \approx 0.9$
 $\log_9 5 \approx 0.7$
Find $\log_9 324$